KEGS 2026
SYMPOSIUM

A 3D hierarchical finite-volume approach for time-domain electromagnetic simulation with thin, highly conductive structures

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CANADIAN EXPLORATION
GEOPHYSICAL SOCIETY

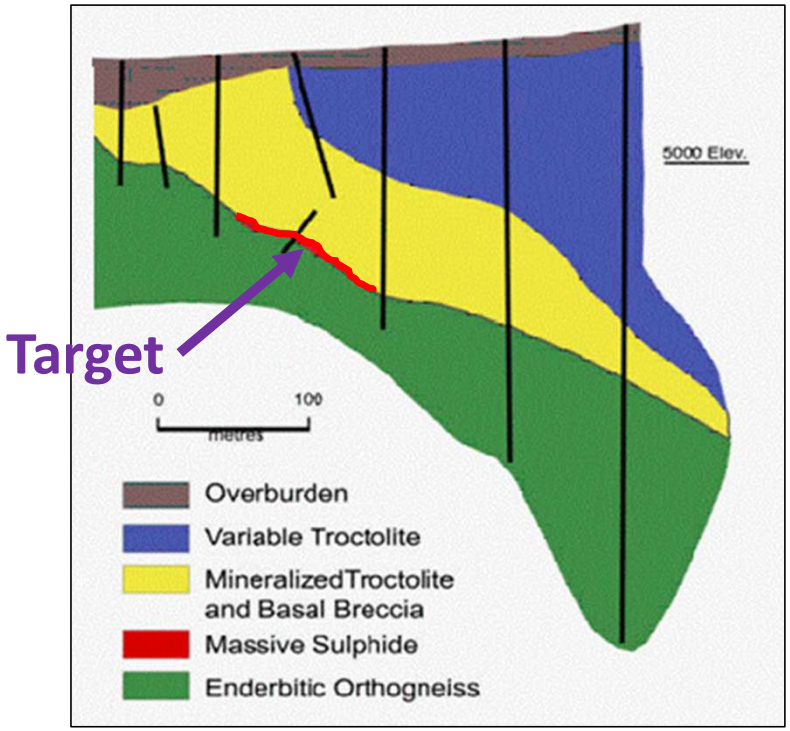
Presentation Outline

1. Motivation
2. Hierarchical Finite Volume Method
3. Modeling Studies

1. Motivation

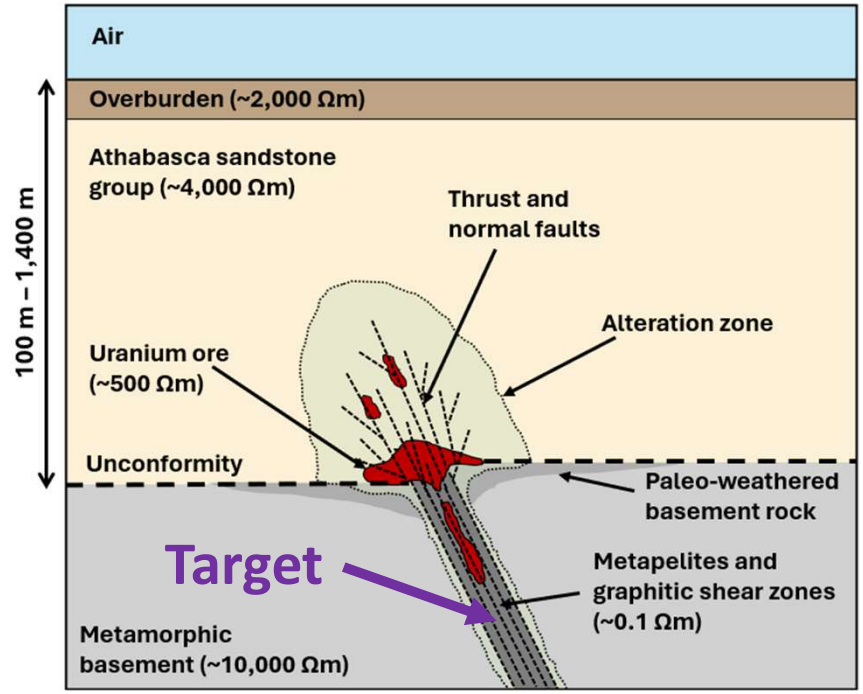
Sheet-Like Conductor in a Complex Host

Massive Nickel-Sulphide Deposits



Modified from Independent Technical Report - Voisey's Bay Project Royalty, Labrador, Canada by Gustavson Associates, 2005.

Unconformity Uranium



EM Methods

- Infer target structure from field collected geophysical data

  → Hypothesis testing

  → Geophysical inversion

- Requires effective simulation of EM data

  → Evaluate hypotheses

  → Evaluate inversion models (data misfit)

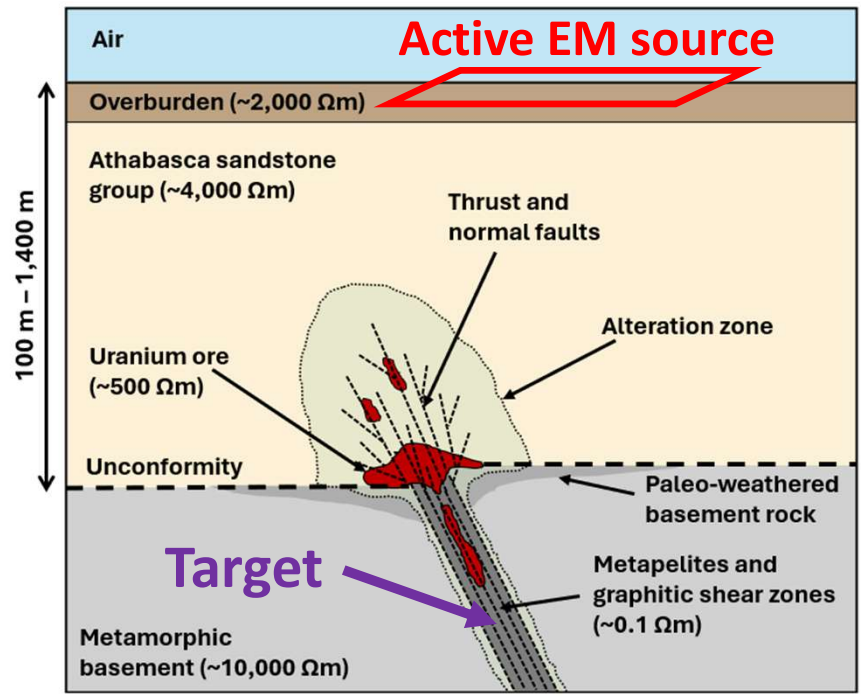


Plate-Modeling

Approach

Fit data signatures with plates

→ Plot agglomeration of plates

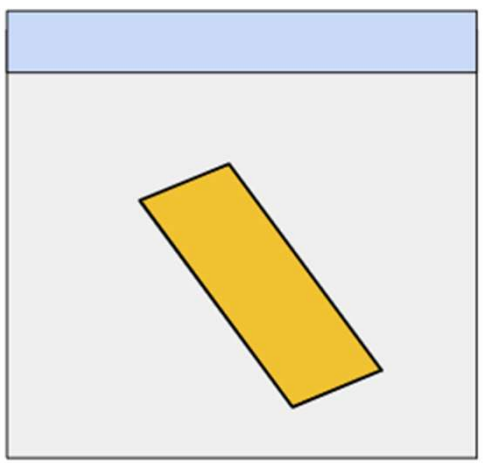
→ Interpret larger structure

Issues

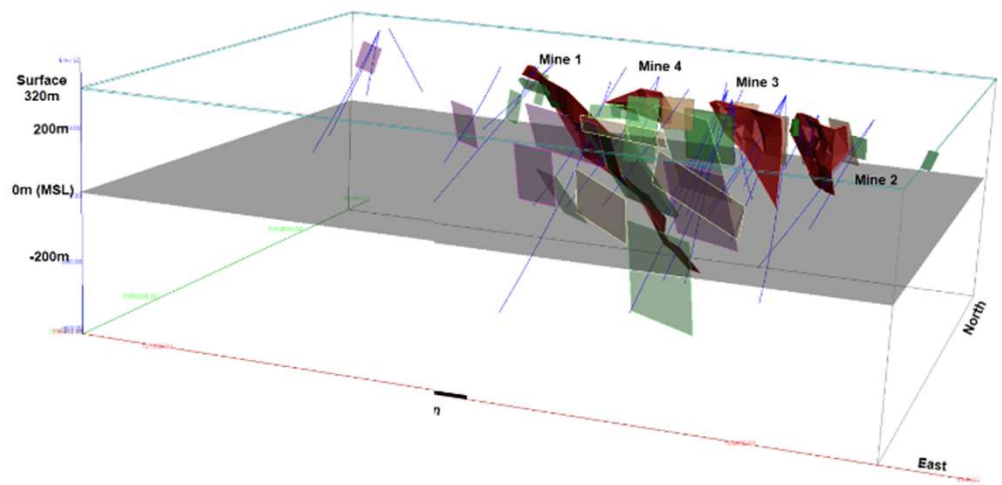
- Simple host and target geometry

  → not geologically realistic
- Does not fit data globally
- Unstable for extreme conductances?

Individual plate model



Agglomerated plate model



Finite Volume Method

Maxwell's Equations

$$\nabla \times \vec{e} + \frac{\partial \vec{b}}{\partial t} = -\vec{s}_m$$

$$\nabla \times \vec{h} - \vec{j} = \vec{s}_e$$

$$\vec{j} = \sigma \vec{e}$$

$$\vec{h} = \mu^{-1} \vec{b}$$

FVM

- Inner products
- Design a mesh
- σ for each cell
- Discretize

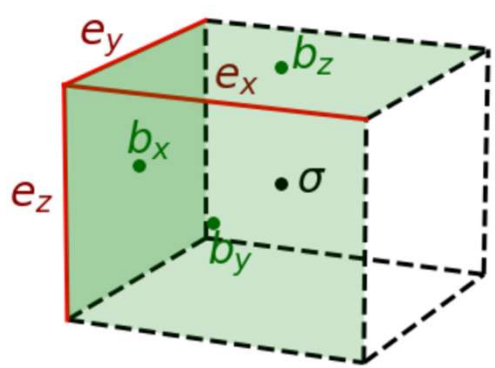
Solvable Linear System

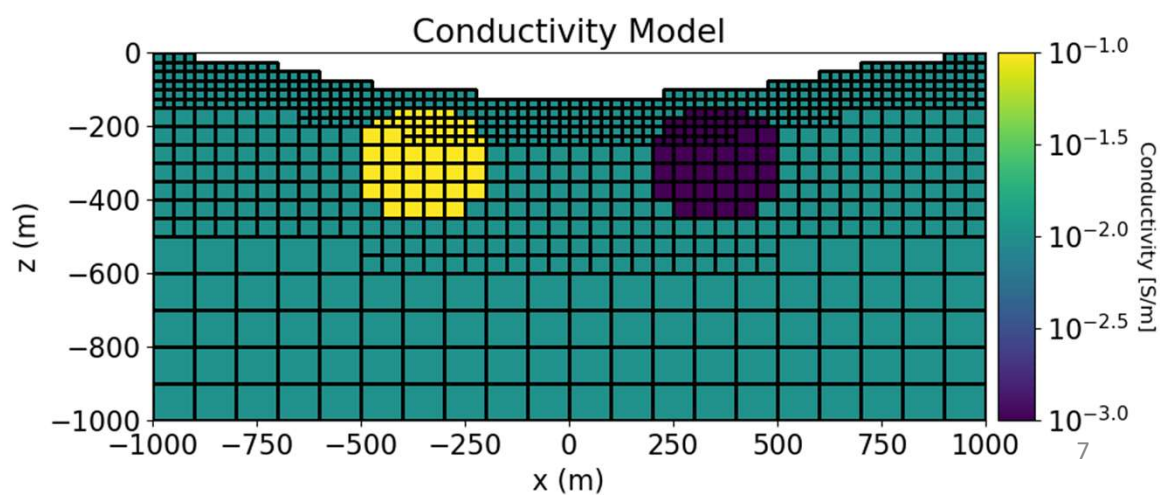
$$\mathbf{C^T M_{f\frac{1}{\mu}} C e} + \mathbf{M_{e\sigma}} \frac{\partial \mathbf{e}}{\partial t} = -\mathbf{C^T M_{f\frac{1}{\mu}} s_m} - \frac{\partial \mathbf{s_e}}{\partial t}$$

OR

$$\mathbf{C M_{e\sigma}^{-1} C^T M_{f\frac{1}{\mu}} b} + \frac{\partial \mathbf{b}}{\partial t} = \mathbf{C M_{e\sigma}^{-1} s_e} - \frac{\partial \mathbf{s_m}}{\partial t}$$

Discretization of fields and fluxes





Finite Volume Method

Accuracy of the forward problem

- Objects at least several cells thick
- Minimum cell size depends on diffusion distance

$$\Delta h < \frac{D}{2} \approx 625\sqrt{\frac{t}{\sigma}}$$

Voxel-based not adept for:

- Highly conductive structures
- Plate-like or curve-like structures

Finite Volume Method

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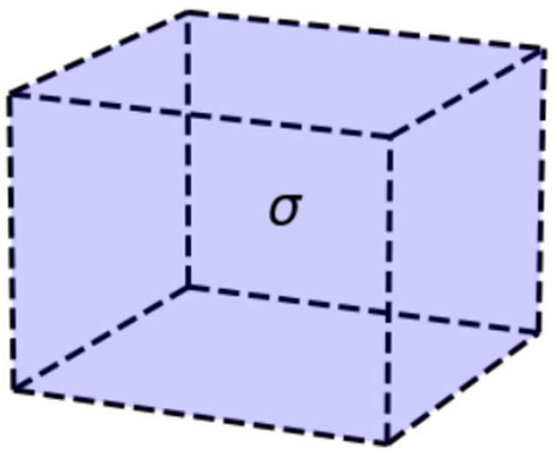
I am motivated to rectify this

2. Hierarchical Finite Volume Method

Hierarchical Framework (Weiss, 2017)

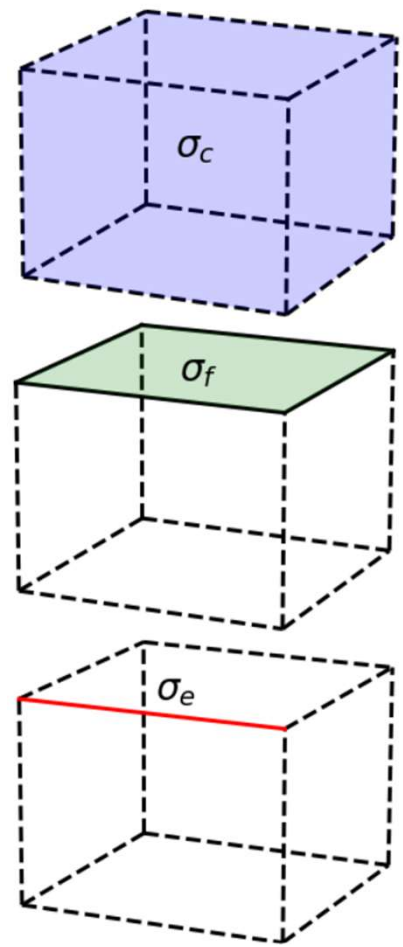
Original Formulation



Ohm's law for cell i:

$$\vec{j} = \sigma_i \vec{e}$$

Hierarchical Framework



We discretize:

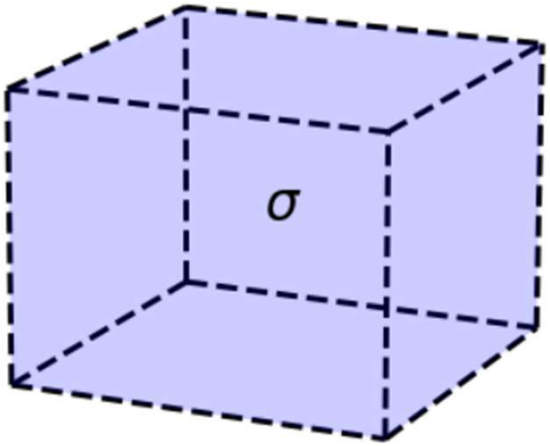
- Thick structures to centers
- Plates to faces
- Curves to edges

Distinct conductivity:

- Within cells
- On faces
- On edges

Hierarchical Finite Volume

Original Formulation



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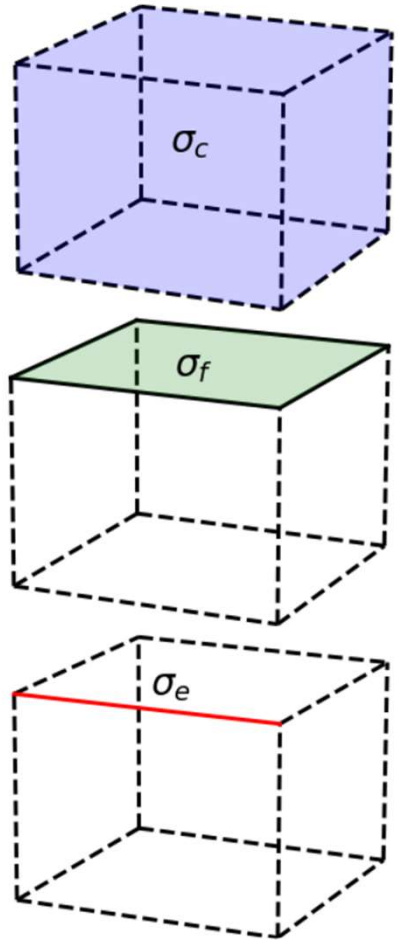


Volume currents within cell i:

$$\vec{j} = \sigma_i \vec{e}$$

Surface currents on face i:

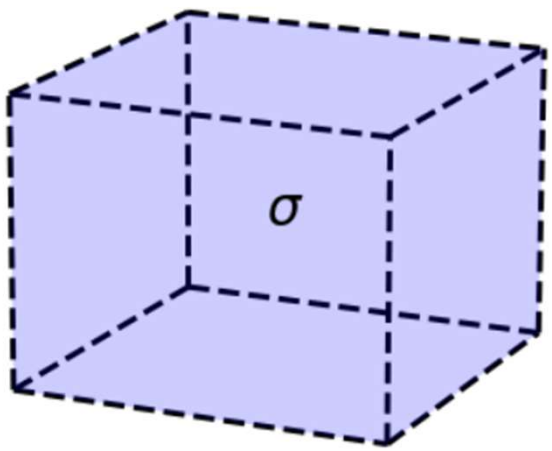
$$\vec{K} = \tau_i \vec{e}$$

Line currents along edge i:

$$\vec{\ell} = \lambda_i \vec{e}$$

Hierarchical Finite Volume

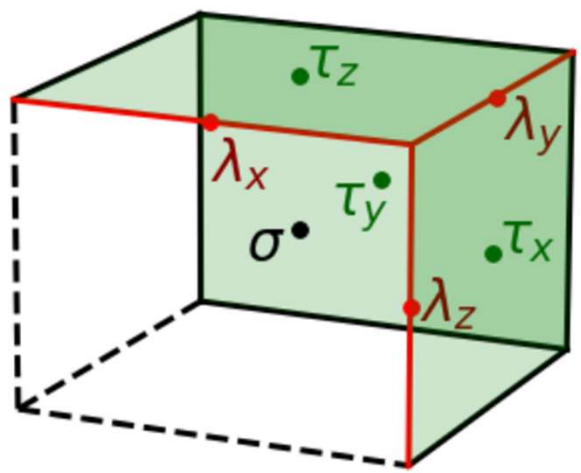
Original Formulation



$$\int_V \vec{u} \cdot \vec{j}_f \, dv = \int_V \vec{u} \cdot \sigma \vec{e} \, dv$$

$$\mathbf{M_e\, j} = \mathbf{M_{e\sigma}\, e}$$

$$\int_V \vec{u} \cdot \vec{j}_f \, dv = \sum_i^{nC} \int_{V_i} \vec{u} \cdot \sigma_i \vec{e} \, dv + \sum_i^{nF} \int_{A_i} \vec{u} \cdot \tau_i \vec{e} \, da + \sum_i^{nE} \int_{L_i} \vec{u} \cdot \lambda_i \vec{e} \, dl$$



$$\mathbf{M_e\, j} = \left(\mathbf{M_{e\sigma}} + \mathbf{M_{e\tau}} + \mathbf{M_{e\lambda}}\right) \mathbf{e}$$

Hierarchical Finite Volume

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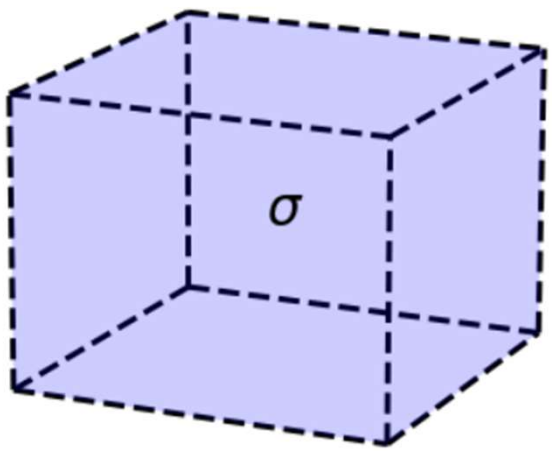

$$\int_V \vec{u} \cdot \vec{j}_f \, dv = \int_V \vec{u} \cdot \sigma \vec{e} \, dv$$

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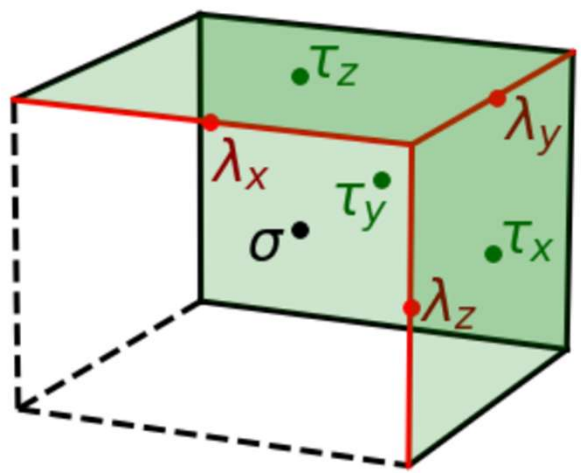
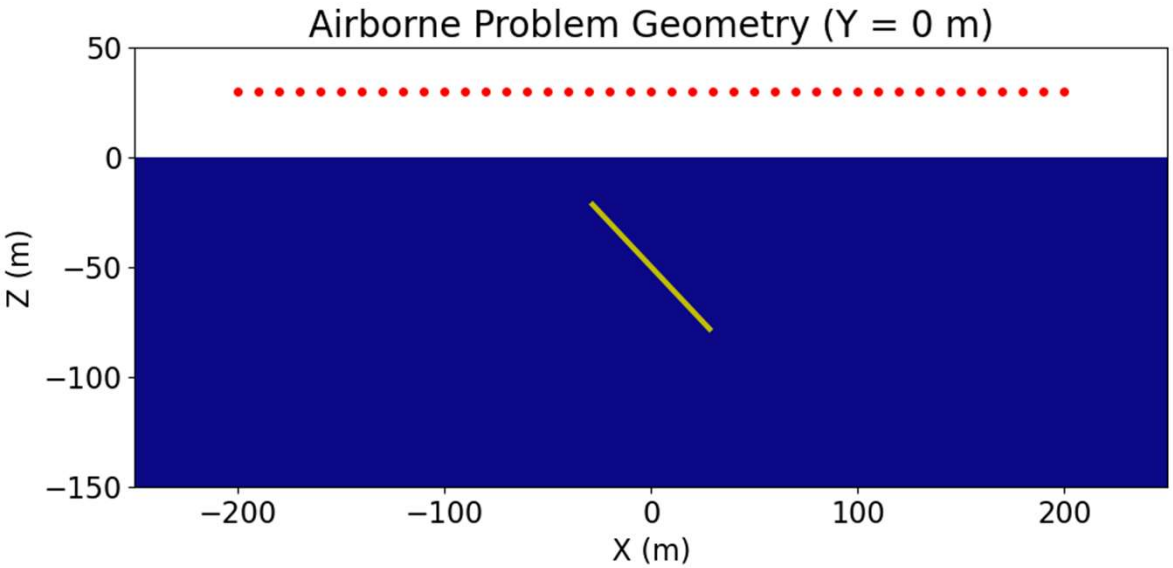


$$\mathbf{M_e\, j} = \boxed{\left(\mathbf{M}_{\mathbf{e}\sigma} + \mathbf{M}_{\mathbf{e}\tau} + \mathbf{M}_{\mathbf{e}\lambda}\right)}\, \mathbf{e}$$

3. Modeling Studies

Airborne TDEM Example

- Simulate b(t) and db/dt for coincident loop airborne survey line
- Square pulse waveform (0.01 s on-time)
- Thin dipping rectangular conductor in a 1e-4 S/m host



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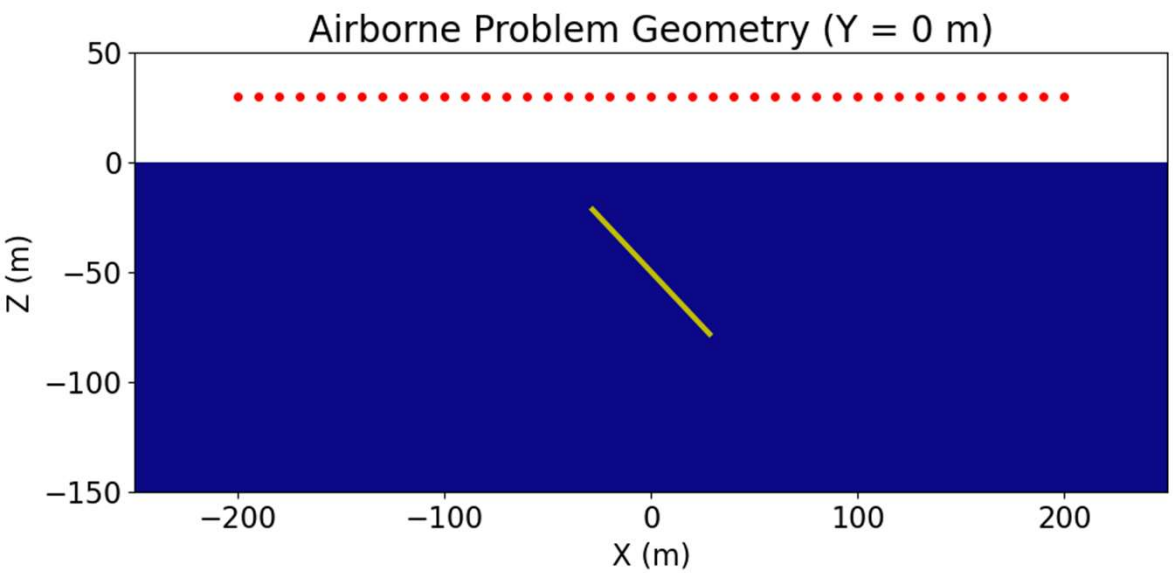
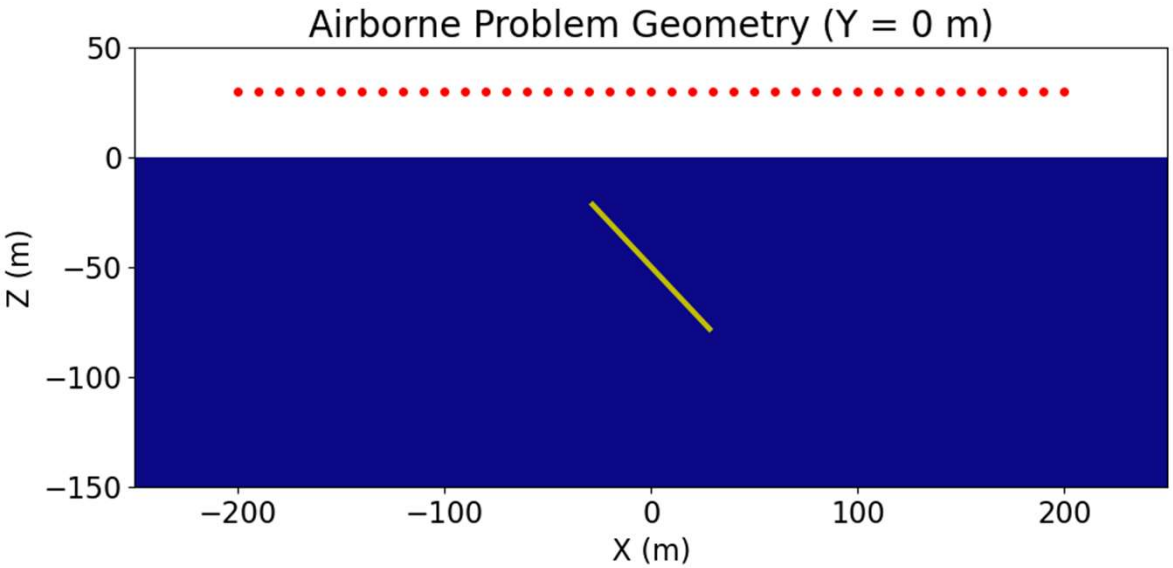


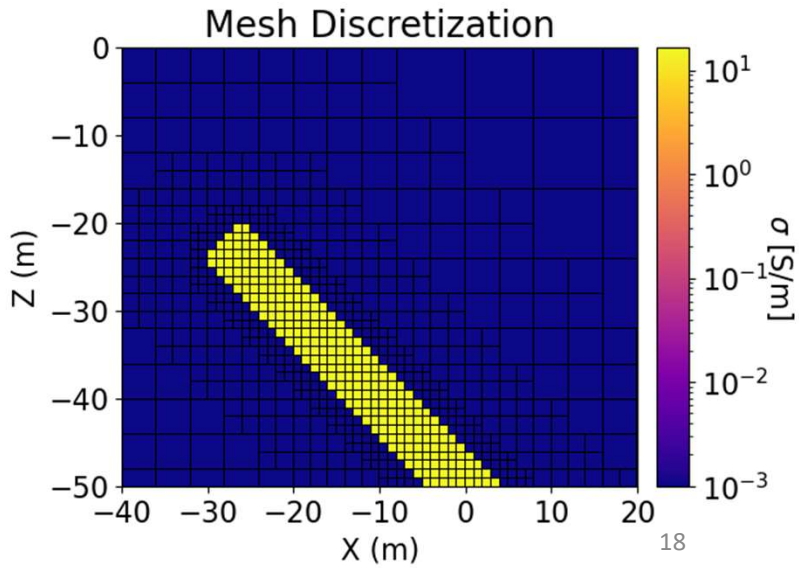
Compare 3 methods:

- Standard finite volume
- LEROI plate modeling
- Hierarchical finite volume

Method 1: Standard Finite Volume

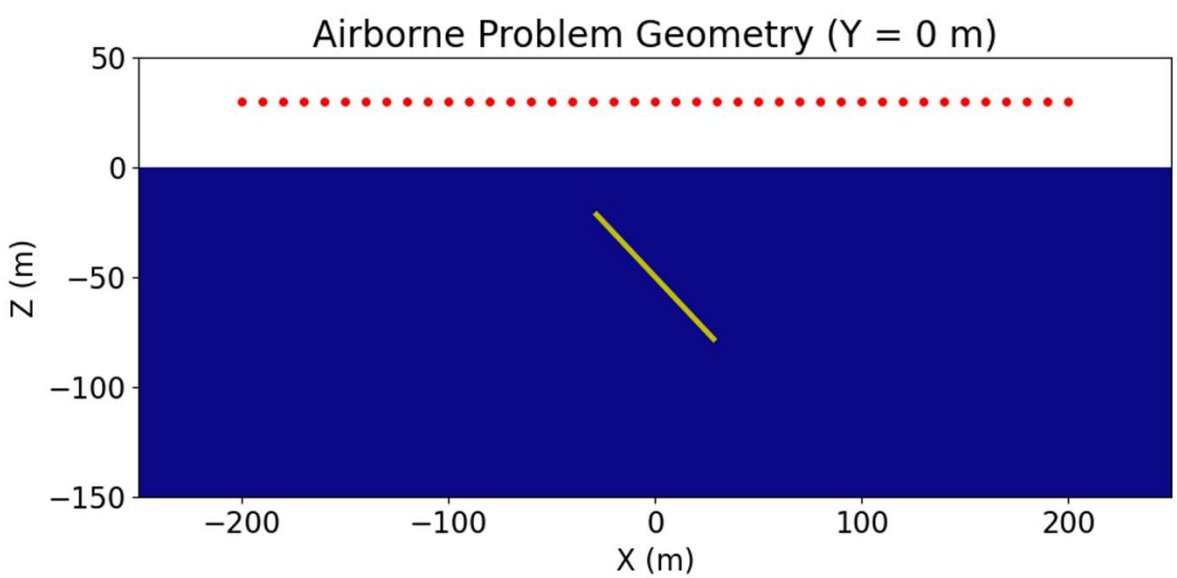
- Target is 6 m thick and 16.7 S/m (100 S)
- 1 m min cell size → 153,889 mesh cells
- >32 GB RAM, 20 min on a cluster



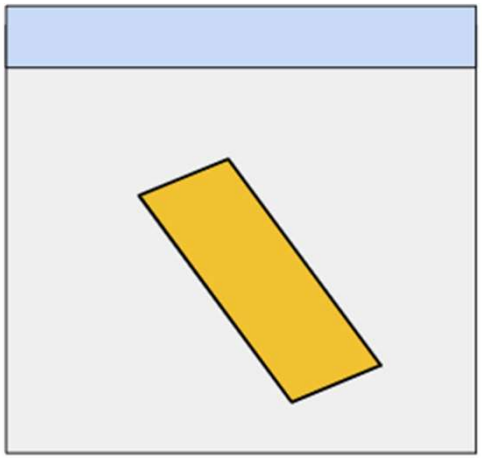


Method 2: P223 LEROI

- 100 S plate in a 1e-4 S/m halfspace
- Integral formulation
- Negligible RAM, roughly 1 min to compute

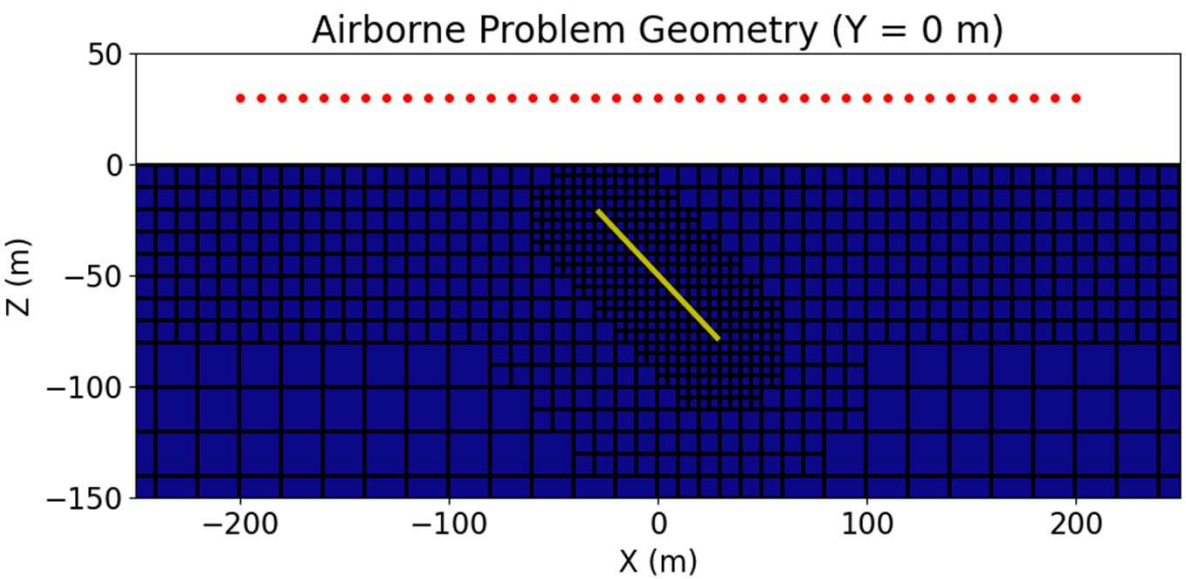


Individual plate model

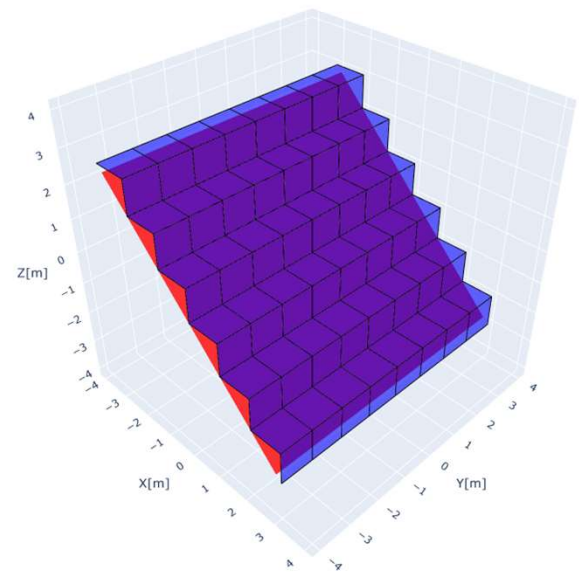


Method 3: Hierarchical Finite Volume

- 100 S plate within a 1e-4 S/m halfspace
- Step-projection → 100 S conductances on mesh faces
- 5 m min cell size → 58,512 mesh cells
- <8 GB RAM, 5 min on a laptop

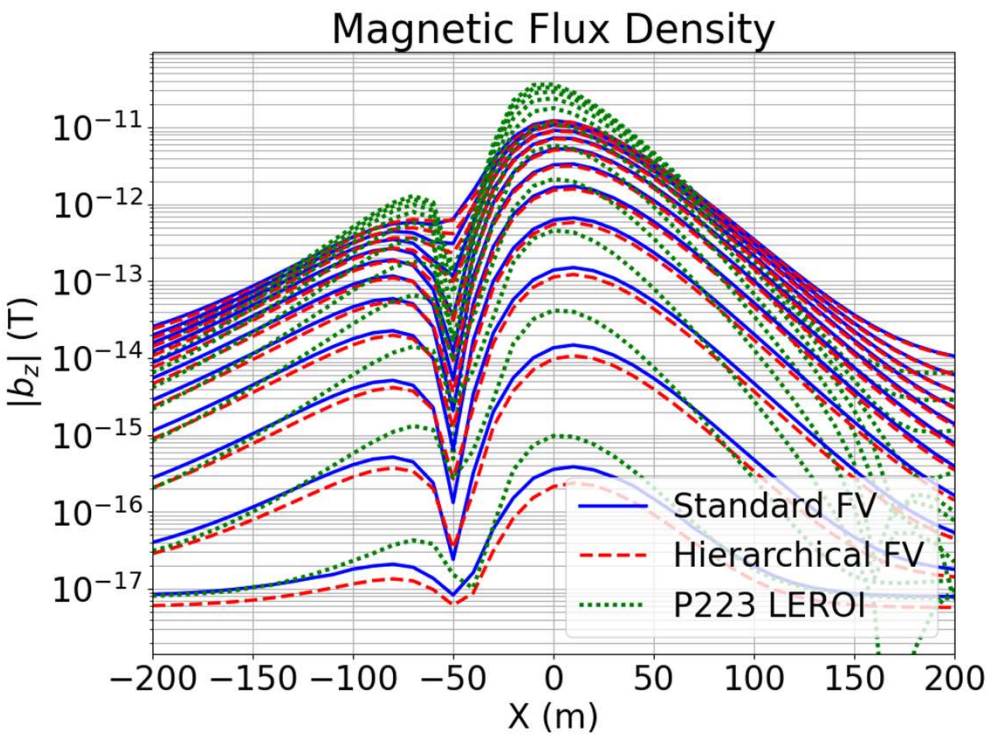


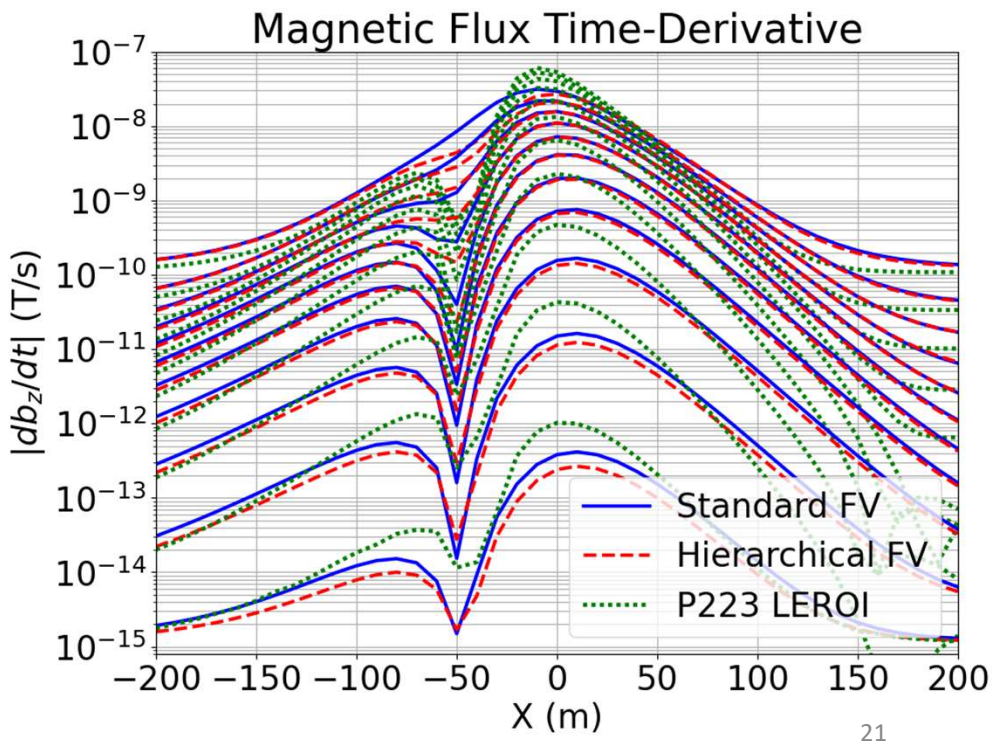
Step projection



Code Comparison

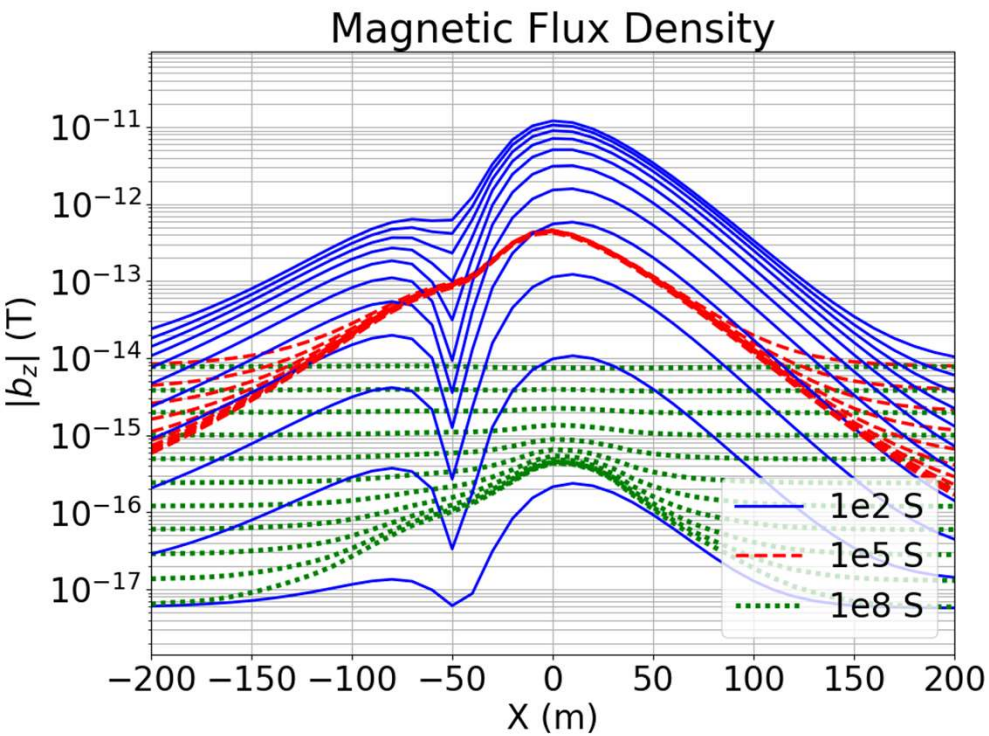
- Hierarchical FV better captures physics than plate modeling
- Hierarchical FV requires fraction of computational resources of standard FV

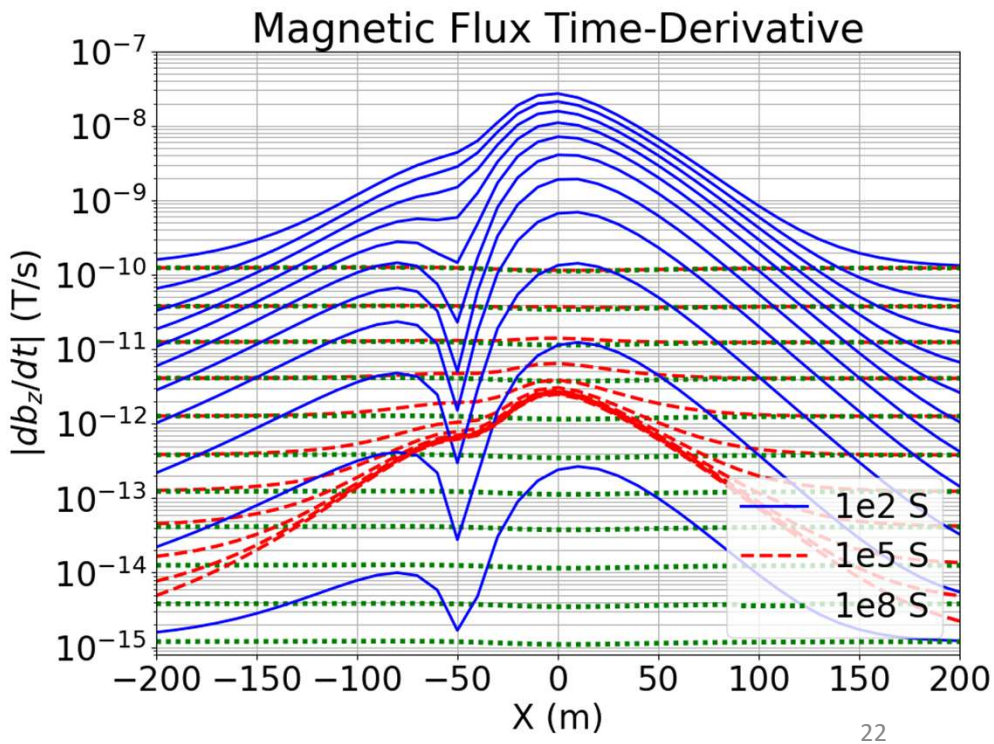




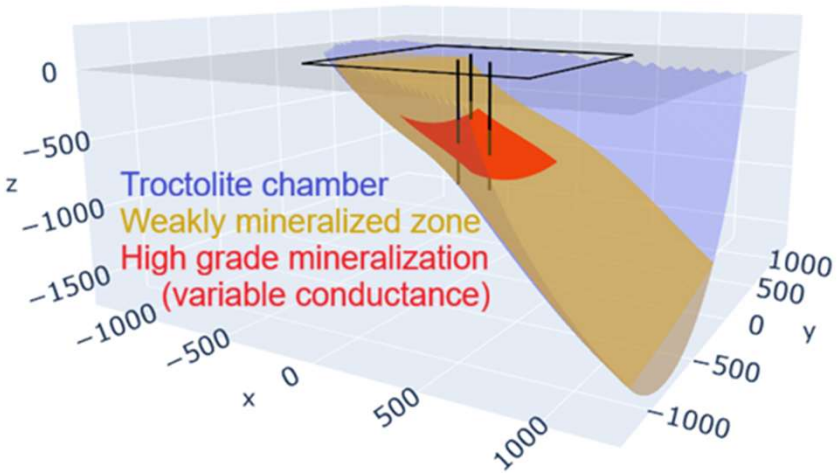
Hierarchical FV for High Conductances

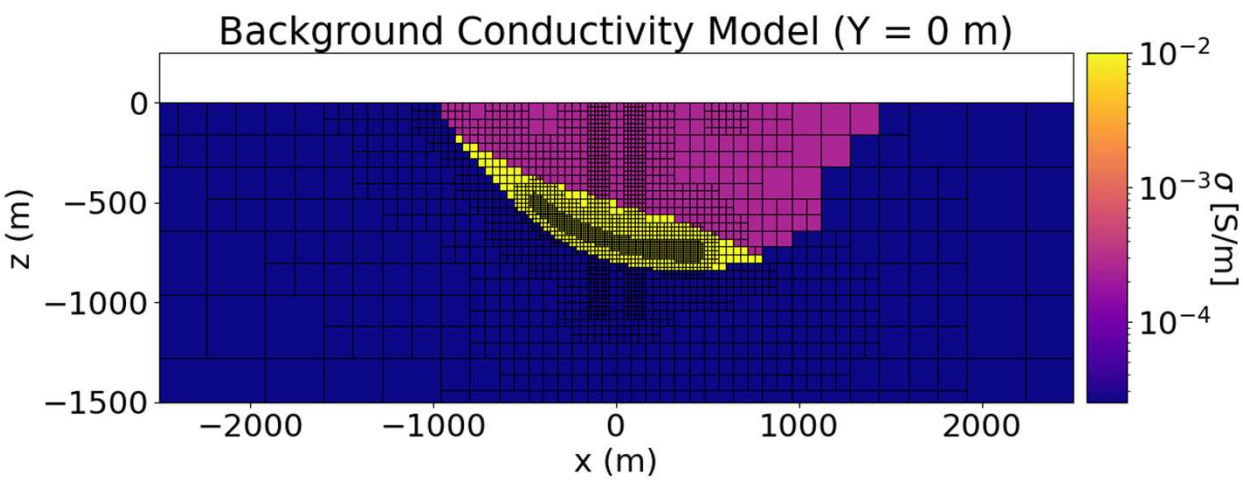
- Simulate for 1e2 S, 1e5 S and 1e8 S conductances
- Exhibits stability and expected data signatures





Voisey's Bay Inspired Example

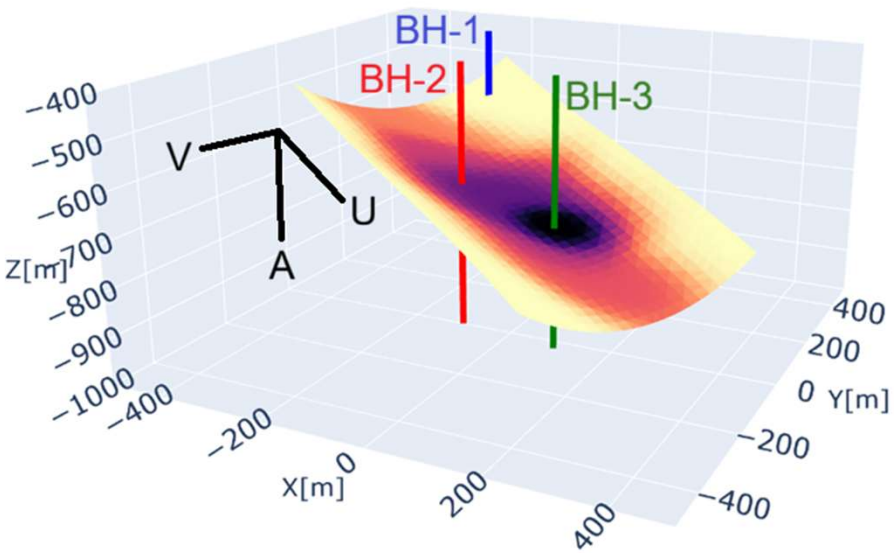




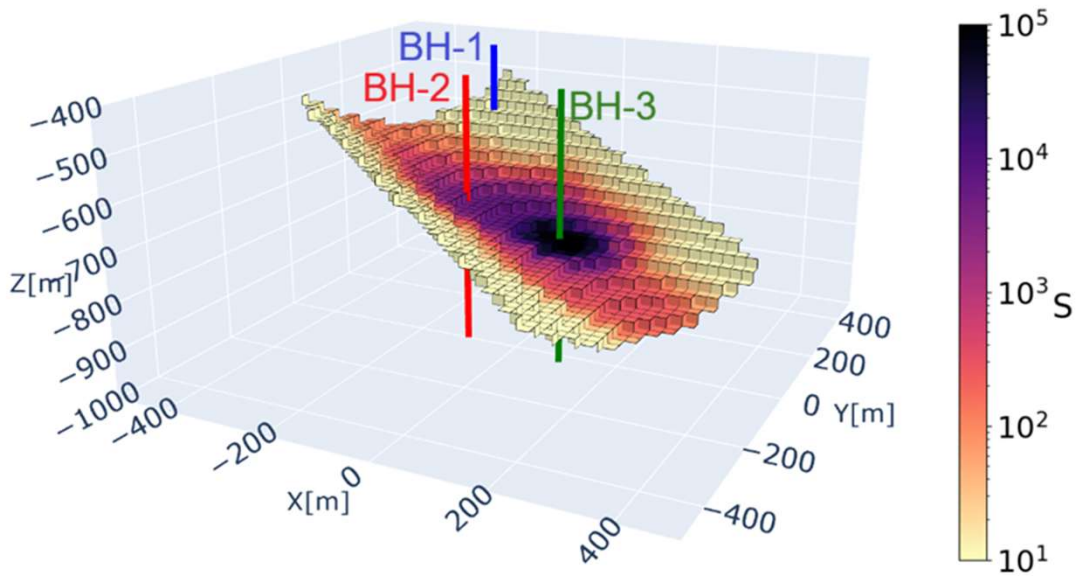
- Simulate Borehole UTEM data
- Background conductivity model:
  - Orthogneiss (2.5e-5 S/m)
  - Troctolite (2.5e-4 S/m)
  - Disseminated zone (1e-2 S/m)
- High-grade ore characterized by a sheet-like conductor (requires conductance model)

Voisey's Bay Inspired Example

Delaunay conductance model

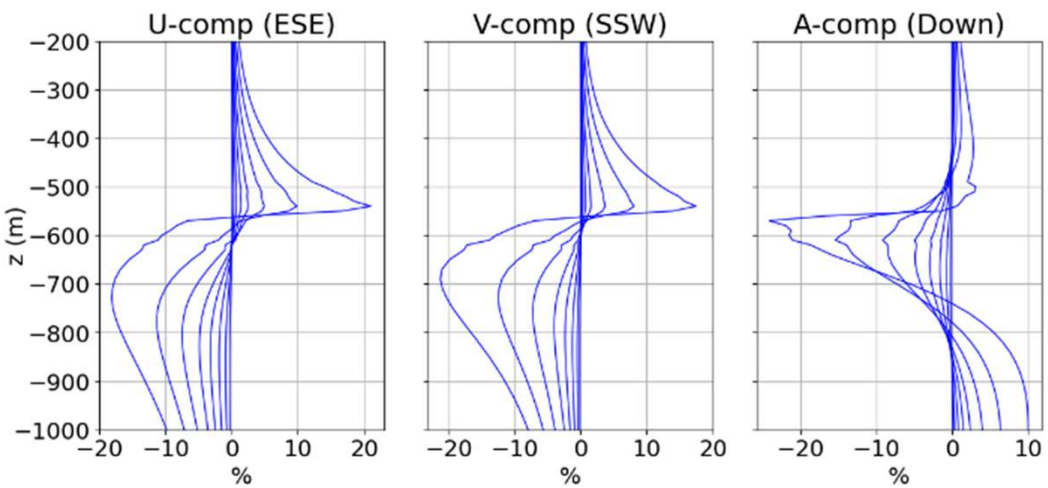


Conductances mapped to mesh faces



- Define conductance for each triangle in a 3D Delaunay surface (model)
- Project to OcTree mesh (linear mapping)

Channel Reduced Data: BH-1



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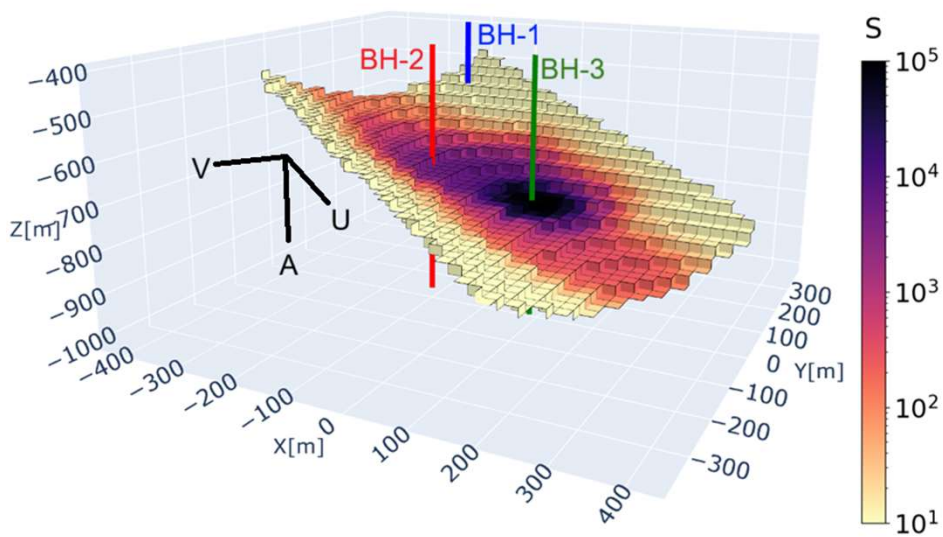
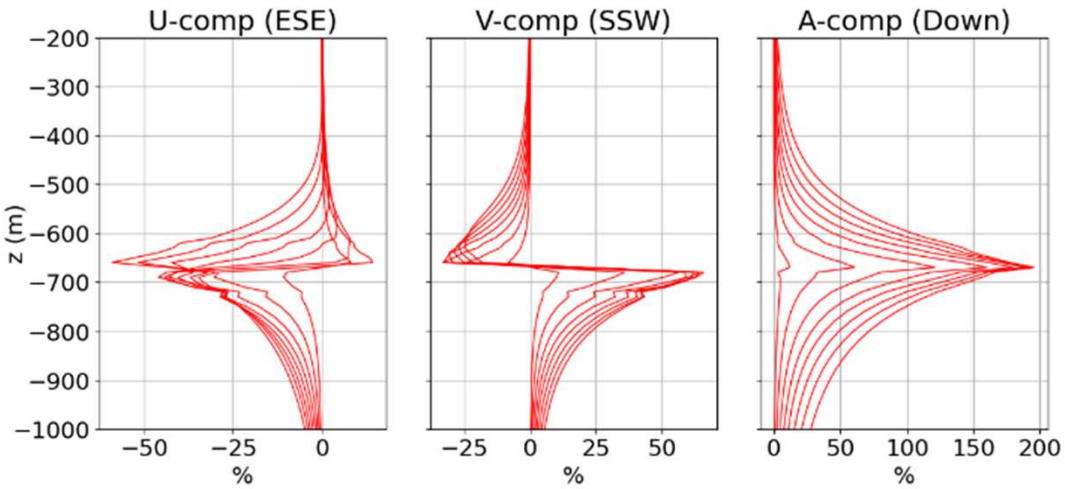
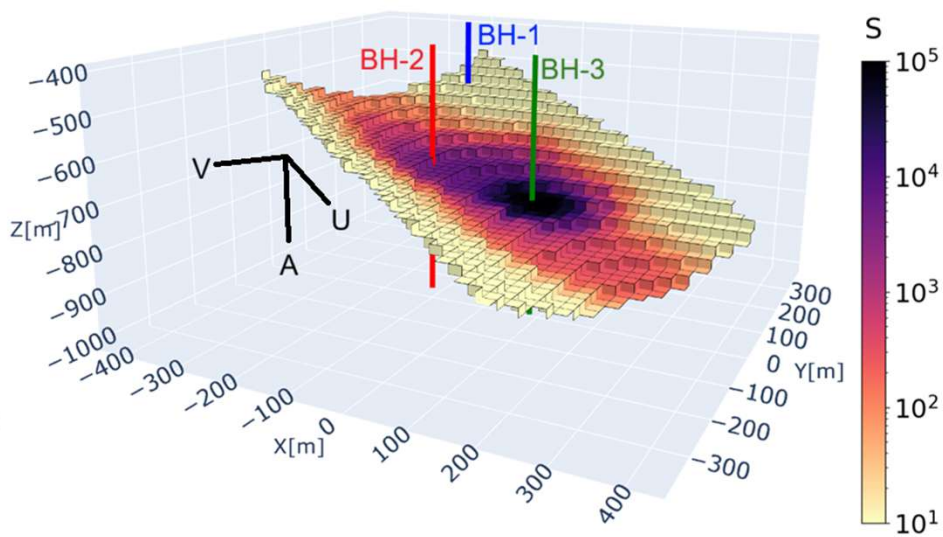


- Outside the dominant conductor
- Dominant conductor offset towards along-line (U) and cross-line (V) direction

Channel Reduced Data: BH-2

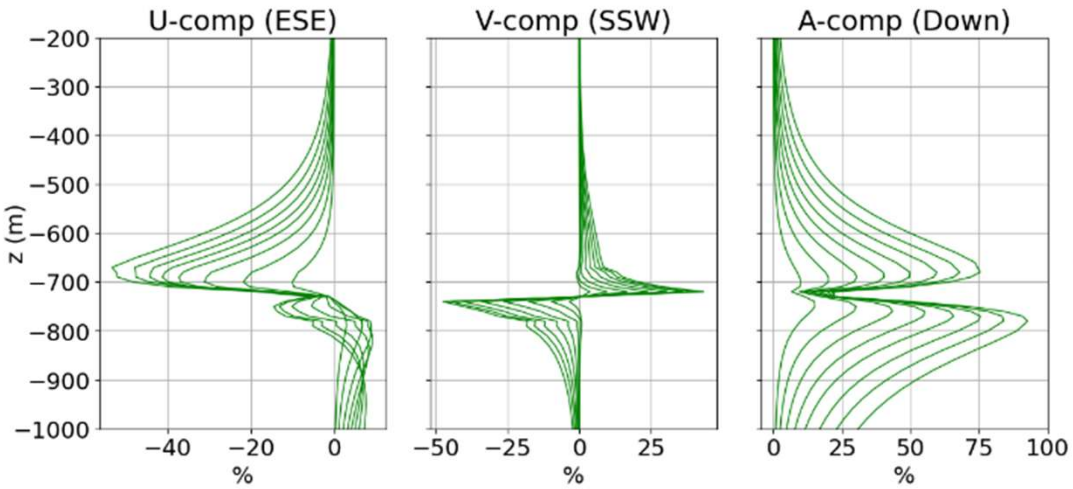


Conductances mapped to mesh faces

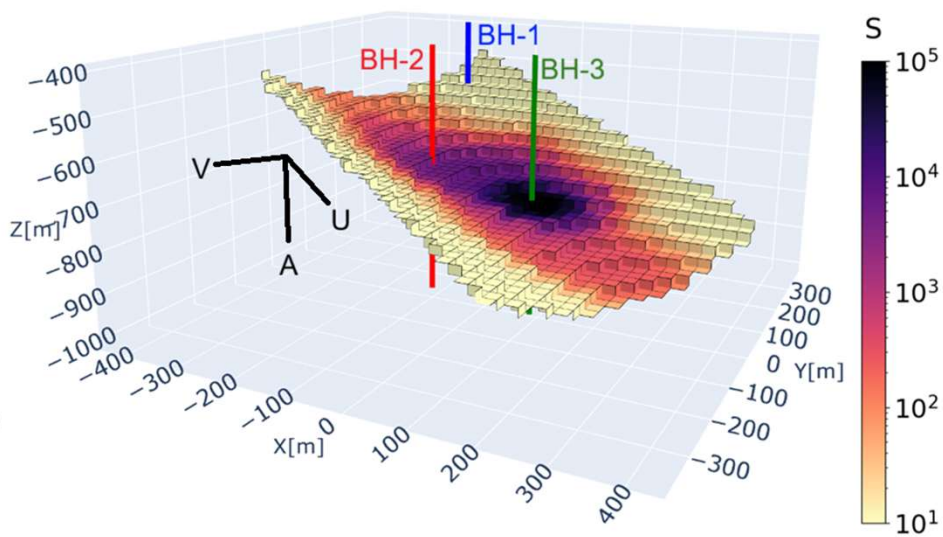


- Intersection with moderate conductance at ~675 m

Channel Reduced Data: BH-3

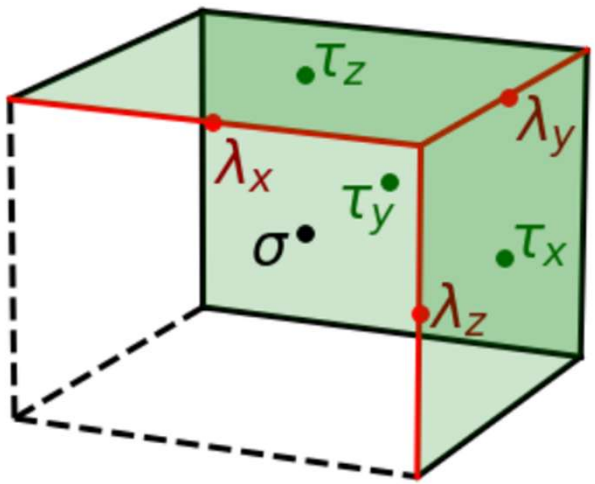


Conductances mapped to mesh faces



- Intersection with high conductance at ~725 m

Summary



- Hierarchical finite volume parameterizes electrical properties to cell volumes, faces (and edges)
- More complete physics than plate-modeling with less resources than voxel approaches
- Stable for high conductance plates (>1e5 S)
- Geometric complexity and inhomogeneity in the host and target
- Computationally tenable for testing complex hypotheses
- Built using the open-source framework SimPEG

THANK YOU